

Lightweight Storage and Offline Reconstruction of Soccer Tracking Data

DongGwon Kang, GeonHo Kim (Team Name: doopdoop)

Abstract—Player tracking data is widely used in modern soccer analytics, but sampling, storing, and processing high-frequency trajectories can impose substantial storage, I/O, and computational overhead in large-scale data pipelines. In this paper, we consider an original tracking stream sampled every 0.04 s as the ground truth and investigate a lightweight storage scheme that stores trajectories at 0.08 s intervals. This reduced-rate representation decreases the number of stored frames, while the discarded intermediate samples are reconstructed offline when higher-rate input is required.

We consider two reconstruction methods for the removed intermediate frames: a linear-interpolation baseline and an LSTM-based predictor. The linear method estimates the missing midpoint between two retained samples under a constant-velocity assumption, while the LSTM-based method uses local motion context to predict the discarded intermediate position. Both reconstructed outputs are evaluated against the original 0.04 s ground-truth data.

The goal of this work is not to generate artificially enhanced tracking data, but to examine whether reduced-rate storage can preserve the essential statistical and kinematic characteristics of the original signal. All preprocessing, downsampling, reconstruction, and evaluation procedures are designed for an offline Hadoop-based pipeline, making the proposed scheme compatible with large-scale sports-data infrastructures. This framework provides a practical basis for analyzing the trade-off between tracking-data fidelity and infrastructure cost in soccer analytics.

Index Terms—soccer analytics, player tracking, trajectory interpolation, LSTM, Hadoop, downsampling, sports big data.

I. INTRODUCTION

Player tracking data is widely used in modern soccer analytics for tactical analysis, physical-load monitoring, player evaluation, and event-based modeling. High-frequency tracking data can capture detailed player and ball movements, but sampling, storing, and processing every frame creates substantial overhead in large-scale analytics pipelines. This cost becomes especially important when many matches or full-season datasets are managed in distributed storage and batch-processing environments such as HDFS and Hadoop.

In this paper, we consider an original tracking stream sampled every 0.04 s as the ground truth. To reduce storage and processing cost, the data are downsampled and stored at 0.08 s intervals. This halves the number of stored frames, but removes the intermediate samples that may be needed for downstream analysis. Therefore, the main question of this work is whether the discarded 0.04 s intermediate frames can be reconstructed accurately from the 0.08 s stored data.

We evaluate two reconstruction methods. The first is linear interpolation, which estimates the missing midpoint between two retained samples under a constant-velocity assumption. The second is an LSTM-based predictor, which uses nearby retained samples to infer the missing intermediate position

from local motion context. Both reconstructed outputs are compared against the original 0.04 s ground-truth data.

The goal of this study is not to generate artificially enhanced tracking data, but to determine whether reduced-rate storage can preserve the essential statistical and kinematic characteristics of the original signal. If the reconstruction error remains sufficiently small, storing tracking data at 0.08 s intervals and reconstructing missing frames offline can provide a practical lightweight alternative to storing the full 0.04 s stream.

The contributions of this paper are summarized as follows:

- We formulate a lightweight tracking-data scheme that stores soccer trajectories at 0.08 s intervals while using the original 0.04 s data as ground truth for evaluation.
- We compare two intermediate-frame reconstruction methods, linear interpolation and LSTM-based prediction, against the original 0.04 s ground-truth samples.
- We implement the downsampling, reconstruction, and evaluation process in a Hadoop-based offline pipeline, showing that the proposed scheme can be integrated into large-scale sports-data infrastructures.

II. BACKGROUND AND MOTIVATION

A. Soccer Player Tracking Data

Soccer player tracking data describe the movement of players and the ball over time. Each frame contains two-dimensional pitch coordinates, where a player position at time t is represented as

$$\mathbf{p}_t = (x_t, y_t)^\top.$$

By connecting these positions over time, a player trajectory is obtained. In this work, the original tracking stream is sampled every 0.04 s and is used as the ground-truth high-frequency trajectory.

B. Storage and Processing Cost

Although 0.04 s tracking data provide detailed movement information, they produce a large number of frames. As the number of matches increases, this leads to higher storage usage, larger I/O cost, and longer batch-processing time in large-scale platforms such as HDFS and Hadoop. Therefore, storing tracking data at a lower temporal resolution can reduce infrastructure cost while keeping the main movement patterns.

C. Intermediate-Frame Reconstruction

When the original 0.04 s data are downsampled to 0.08 s intervals, every other frame is removed. However, some downstream analyses may still require the original temporal resolution. To address this, the removed intermediate frames can be reconstructed offline from the retained samples.

D. Linear Interpolation

Linear interpolation is a simple reconstruction method that estimates the missing intermediate position by assuming constant motion between two retained samples. Given two stored positions before and after the missing frame, the intermediate position can be estimated as their midpoint:

$$\hat{\mathbf{p}}_t = \frac{1}{2} (\mathbf{p}_{t-1} + \mathbf{p}_{t+1}).$$

This method is computationally efficient and does not require model training, making it a useful baseline for evaluating whether more complex prediction models are necessary.

E. LSTM-Based Prediction

Long Short-Term Memory (LSTM) networks are recurrent neural networks designed to model sequential data. In this work, an LSTM-based predictor uses nearby retained trajectory samples as temporal context to estimate the removed intermediate position. Unlike linear interpolation, which assumes constant motion between two points, the LSTM-based method can learn motion patterns from training data and may better handle changes in player speed or direction. Given nearby retained samples as input features, the LSTM encodes local motion context into a hidden state:

$$\mathbf{h}_t = \text{LSTM}(\mathbf{x}_t, \mathbf{h}_{t-1}).$$

The removed intermediate position is predicted by applying a linear output layer to the hidden state:

$$\hat{\mathbf{p}}_t = W_o \mathbf{h}_t + \mathbf{b}_o,$$

where $\hat{\mathbf{p}}_t$ denotes the predicted position and W_o and $\mathbf{b}_o$ are learnable parameters.

III. PROPOSED FRAMEWORK

The proposed framework reduces the storage cost of soccer tracking data by storing trajectories at a lower temporal resolution and reconstructing the removed intermediate frames when needed. The original tracking stream is sampled every 0.04 s and is treated as the ground-truth trajectory. To construct a lightweight representation, every other frame is retained, resulting in a 0.08 s tracking sequence.

Let the original high-frequency trajectory be

$$P = (\mathbf{p}_1, \mathbf{p}_2, \dots, \mathbf{p}_T),$$

where $\mathbf{p}_t = (x_t, y_t)^\top$ denotes the player position at frame t . In this work, the reduced-rate trajectory is obtained by keeping the odd-indexed frames:

$$P_{\text{low}} = (\mathbf{p}_1, \mathbf{p}_3, \mathbf{p}_5, \dots).$$

The even-indexed frames are removed from the stored representation and are used as ground truth during evaluation. Equivalently, the opposite alignment could be used by retaining the even-indexed frames and treating the odd-indexed frames as intermediate samples.

When high-frequency trajectories are required, the missing intermediate positions are reconstructed offline from the retained 0.08 s samples. In this paper, two reconstruction

methods are considered: linear interpolation and LSTM-based prediction. The reconstructed positions are then combined with the retained samples to form an approximated 0.04 s trajectory.

This process is designed for an offline Hadoop-based pipeline. The tracking data are stored in HDFS, and the downsampling, reconstruction, and evaluation steps can be performed as batch jobs. Since reconstruction is conducted offline, both past and future retained samples can be used when estimating the missing intermediate frames.

IV. EVALUATION METHOD

Both the linear interpolation baseline and the LSTM-based model are evaluated against the original 0.04 s ground-truth trajectories on a held-out test split. For each test sequence, the reconstructed trajectory is compared with the corresponding ground-truth trajectory using two metrics: root-mean-square error (RMSE) and discrete Fréchet distance.

The RMSE measures frame-wise positional accuracy and is defined as

$$\text{RMSE}(\hat{P}, P) = \sqrt{\frac{1}{T} \sum_{t=1}^T \|\hat{\mathbf{p}}_t - \mathbf{p}_t\|_2^2},$$

where $P = (\mathbf{p}_1, \dots, \mathbf{p}_T)$ is the ground-truth trajectory and $\hat{P} = (\hat{\mathbf{p}}_1, \dots, \hat{\mathbf{p}}_T)$ is the reconstructed trajectory. RMSE is reported in the same units as the pitch coordinates.

The discrete Fréchet distance measures curve-level similarity between two ordered trajectories. For two trajectories $P = (\mathbf{p}_1, \dots, \mathbf{p}_m)$ and $Q = (\mathbf{q}_1, \dots, \mathbf{q}_n)$, it is defined as

$$d_F(P, Q) = \min_{\sigma \in \mathcal{C}(m, n)} \max_{(i, j) \in \sigma} \|\mathbf{p}_i - \mathbf{q}_j\|_2,$$

where $\mathcal{C}(m, n)$ is the set of monotone couplings between the two trajectories [1]. Unlike RMSE, this metric considers the ordered shape of the trajectory and penalizes large local deviations.

For each test trajectory $\mathcal{T}^{(k)}$ and its reconstruction $\hat{\mathcal{T}}^{(k)}$, both metrics are computed independently for the linear baseline and the LSTM model. Final results are reported as the mean over all N test sequences:

$$\bar{d}_F = \frac{1}{N} \sum_{k=1}^N d_F(\hat{\mathcal{T}}^{(k)}, \mathcal{T}^{(k)}), \quad (1)$$

$$\overline{\text{RMSE}} = \frac{1}{N} \sum_{k=1}^N \text{RMSE}(\hat{\mathcal{T}}^{(k)}, \mathcal{T}^{(k)}). \quad (2)$$

A lower value on both metrics indicates better reconstruction performance. Because RMSE captures pointwise accuracy while d_F captures trajectory shape similarity, the two metrics are interpreted together.

REFERENCES

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